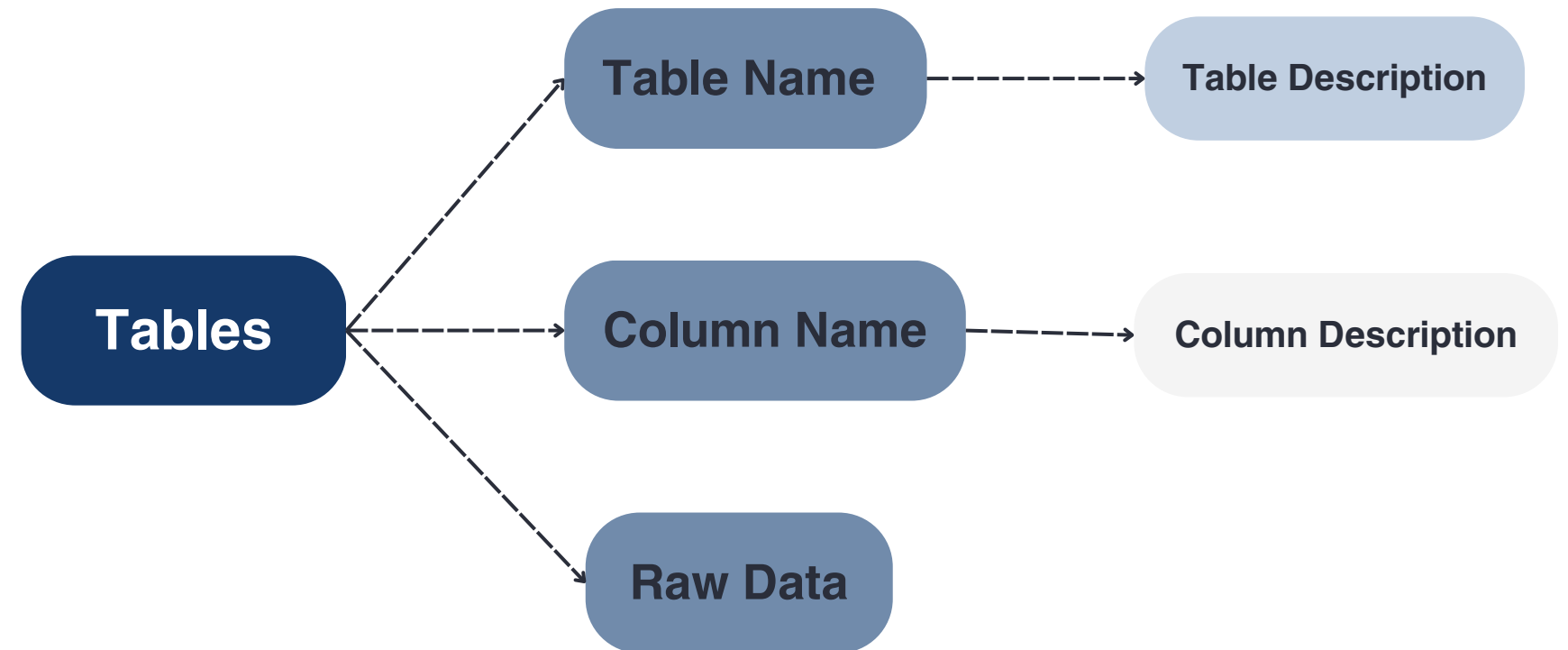


Input Data Structure

In this we talk about the structure of input json file and need for data pre-processing.



Detail About Input Data

Basically, input data is a json file which is consist of 10 tables (products, sales, employee, customer,...) with their description along with column names, column description and raw data related to that table.

Pre-processing data

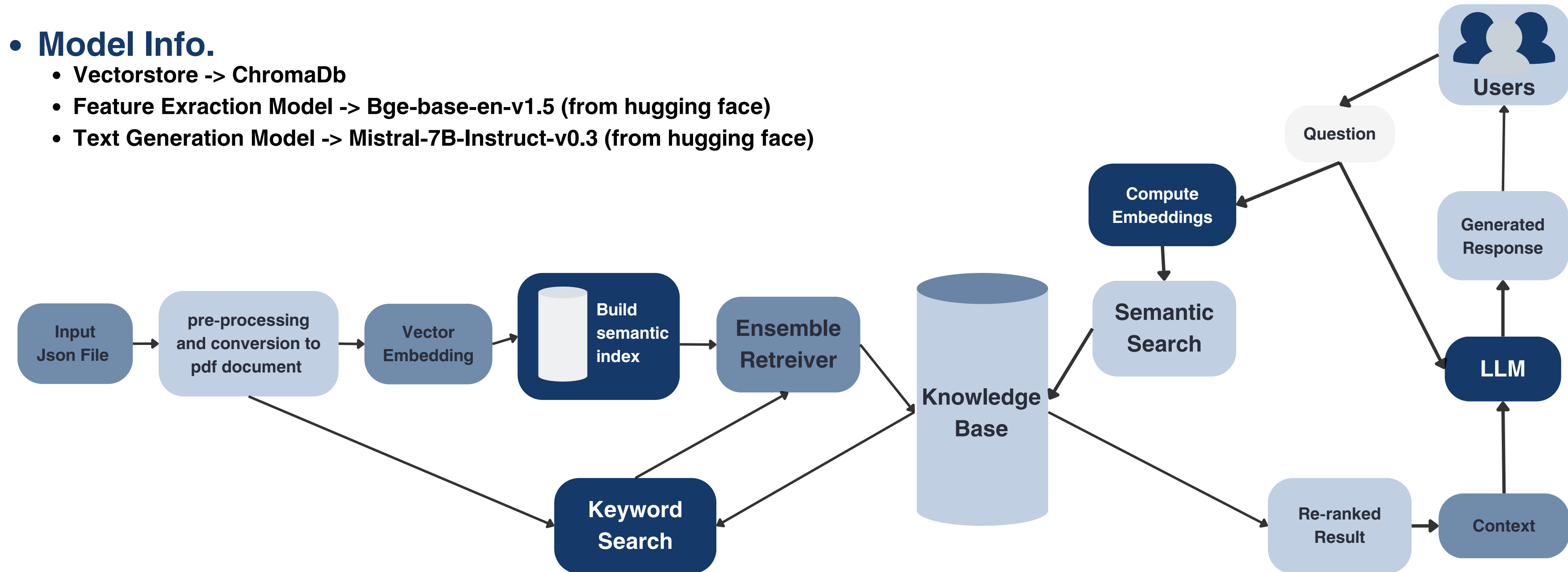
Here we have converted json file to pdf format (unstructure format). In this step we are only deal with table name, table description, column name and column description. This pdf is intermediate data on which we trained LLM model. The thing is we can also give json to model directly but for more accuracy we did this step.

For getting more idea about data set refer this [link](#)

Deep understanding about work flow and algorithim

- **Model Info.**

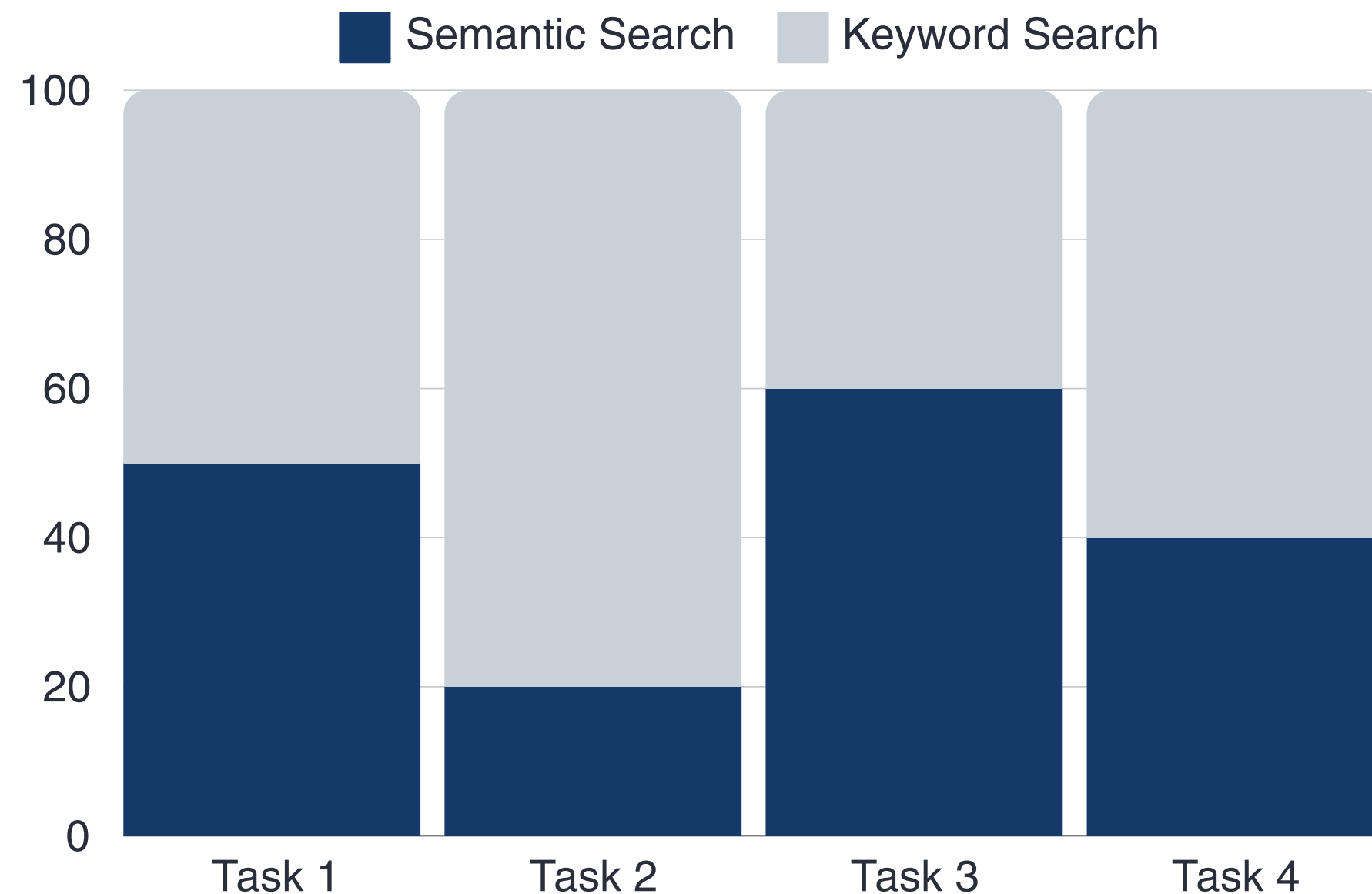
- Vectorstore -> ChromaDb
- Feature Extraxtion Model -> Bge-base-en-v1.5 (from hugging face)
- Text Generation Model -> Mistral-7B-Instruct-v0.3 (from hugging face)



For getting more idea about code refer this [link](#)

Our Purpose

Description of 4 Task



Task 1->SAME DESCRIPTION DIFFERENT COLUMN

We have kept 50 -50 % weightage to both semantic (Will help in getting relationship by description of coulmn) and keyword search (will help in getting relationship by Name of the column)

Task 2 -> SAME COLUMN NO DESCRIPTION

Here we have give 80% weightage to keyword search as there are no descriptiion so the main searching will happen by Column and table name

Task 3-> SAME COLUMN DIFFERENT DESCRIPTION

Here We have given 60% to semantic and 40% to Keyword , So we are just playing around with differtent set to values for getting almost correct answers.

Task 4 -> DIFFERENT COLUMN NO DESCRIPTION

So here We have tested this Case but it is not possible in real Dataset and also our model was also giving some vague answers to queries

OUTPUT OF QUERIES

CASE 1.SAME DESCRIPTION DIFFERENT COLUMN OUTPUT

Query: Give me foreign key relationship with Sales table and other table

1. **buyer_id:** This column in the sales table is a foreign key that references the client_id column in the customers table. This relationship allows us to link a sale to a specific customer.
2. **transaction_product_id:** This column in the sales table is a foreign key that references the product_code column in the products table. This relationship allows us to link a sale to a specific product.
3. **sales_rep:** This column in the sales table is a foreign key that references the employee_id column in the employees table. This relationship allows us to link a sale to a specific sales representative.
4. **delivery_status:** This column in the sales table is a foreign key that references the status column in the delivery_statuses table. This relationship allows us to link a sale to a specific delivery status that each sale involves a product, and the product is stored in the `products` table.

CASE 3. SAME COLUMN DIFFERENT DESCRIPTION OUTPUT

Query: Give me foreign key relationship with department table and other table

1. **`employees` table:** The `manager\_id` column in the `employees` table is a foreign key that references the `employee\_id` column in the `departments` table. This means that each employee has a department manager, and the manager is an employee in the `employees` table.
2. **`departments` table:** The `manager\_id` column in the `departments` table is a foreign key that references the `employee\_id` column in the `employees` table. This means that each department has a manager, who is an employee in the `employees` table.
3. **`sales` table:** The `customer\_id` column in the `sales` table is a foreign key that references the `customer\_id` column in the `customers` table. This means that each sale is made by a customer, and the customer is stored in the `customers` table.
4. **`sales` table:** The `product\_id` column in the `sales` table is a foreign key that references the `product\_id` column in the `products` table. This means that each sale involves a product, and the product is stored in the `products` table.

CASE 2. SAME COLUMN NO DESCRIPTION OUTPUT

Query: Give me foreign key relationship with department table and other table

1. **`employees` table:** The `manager\_id` column in the `employees` table is a foreign key that references the `department\_id` column in the `departments` table. This means that each employee has a department manager, and the department manager is a department in the `departments` table.
2. **`departments` table:** The `location` column in the `departments` table is not a foreign key, but it could be used to establish a relationship with other tables, such as `warehouses` or `stores`, if those tables have a `location` column as well.
3. **`inventory` table:** The `warehouse\_location` column in the `inventory` table could be a foreign key that references the `location` column in the `departments` table, if the warehouse or storage facility is associated with a specific department.
4. **`marketing\_campaigns` table:** The `region` column in the `marketing\_campaigns` table could be a foreign key that references the `location` column in the `departments` table, if the marketing campaigns are targeted based on department locations.

CASE 4. DIFFERENT COLUMN NO DESCRIPTION OUTPUT

Query: Give me foreign key relationship with department table and other table

1. **employees table:**
 - manager_id (in employees table) is a foreign key referencing department_id (in departments table). This establishes a relationship between the employees and departments tables, where an employee's manager is associated with a specific department.
2. **departments table:**
 - No foreign keys are defined in the departments table.
3. **inventory table:**
 - product_id (in inventory table) is a foreign key referencing product_id (in products table). This establishes a relationship between the inventory and products tables, where the inventory records are linked to specific products.
4. **sales table:**
 - product_id (in sales table) is a foreign key referencing product_id (in products table). This establishes a relationship between the sales and products tables, where the sales records are linked to specific products.

For getting more idea about output refer this [link](#)